

24/09/26.

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DevOps

* It is a software development methodology which improves the collaboration b/w developers & operations team.

Epic - Multiple modules into one big module.

Steps in DevOps:

- | | | |
|------------------------------|---|-----|
| 1) plan | } | Dev |
| 2) code | | |
| 3) build | | |
| 4) test | | |
| 5) release | } | Ops |
| 6) operate deploy | | |
| 7) operate | | |
| 8) monitor. | | |

Tools:

- 1) Plan → JIRA
- 2) Code → SCM → Git, Github
- 3) Build → To share project as one file → pytorch
- 4) Test → Selenium, Postman, JUnit.
- 5) Integration (Release) - Jenkins, Github Actions.
- 6) Deploy - Docker
- 7) Operate - Kubernetes
- 8) Monitor - Grafana; CloudWatch.

* DevOps is implemented using various Automation tools.

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Continuous → Automation.

DevOps :-

- ↳ Continuous Development.
- ↳ Continuous Testing.
- ↳ Continuous Integration
- ↳ Continuous Deployment.
- ↳ Continuous Monitoring.

CI/CD Process :-

- 1) Commit Change
- 2) Build
- 3) Test
- 4) Deliver (or)
- 5) Deploy.

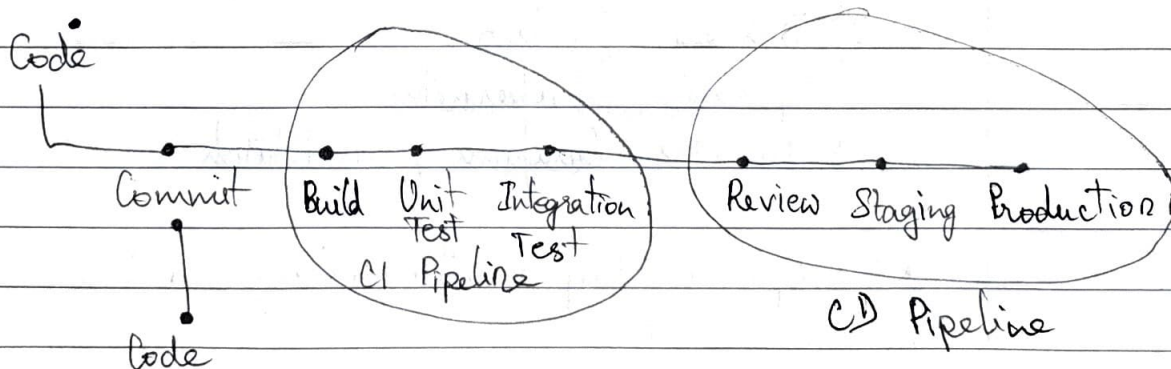
CI Pipeline :-

- 1) Build
- 2) Unit Tests.
- 3) Integration tests.

CD Pipeline :-

- 1) Review
- 2) Staging
- 3) Production.

Automated CI/CD Pipeline :-



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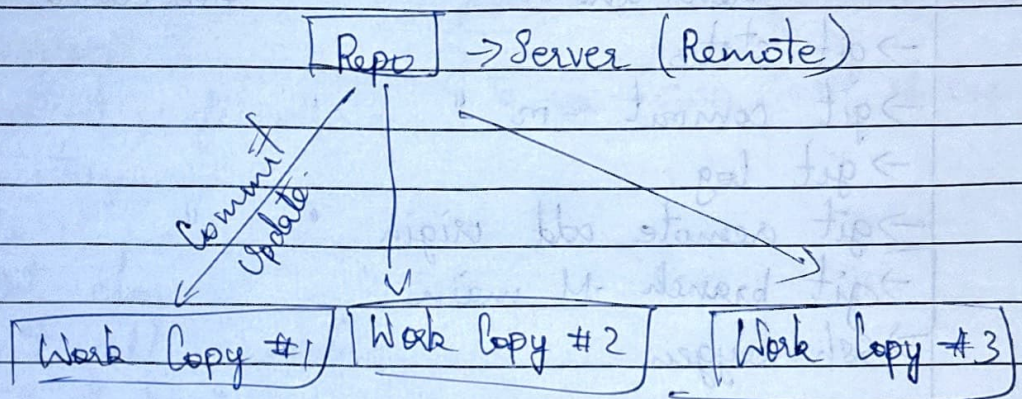
* GitHub → Version Control.

↓
Object Storage.

Source Code Management:

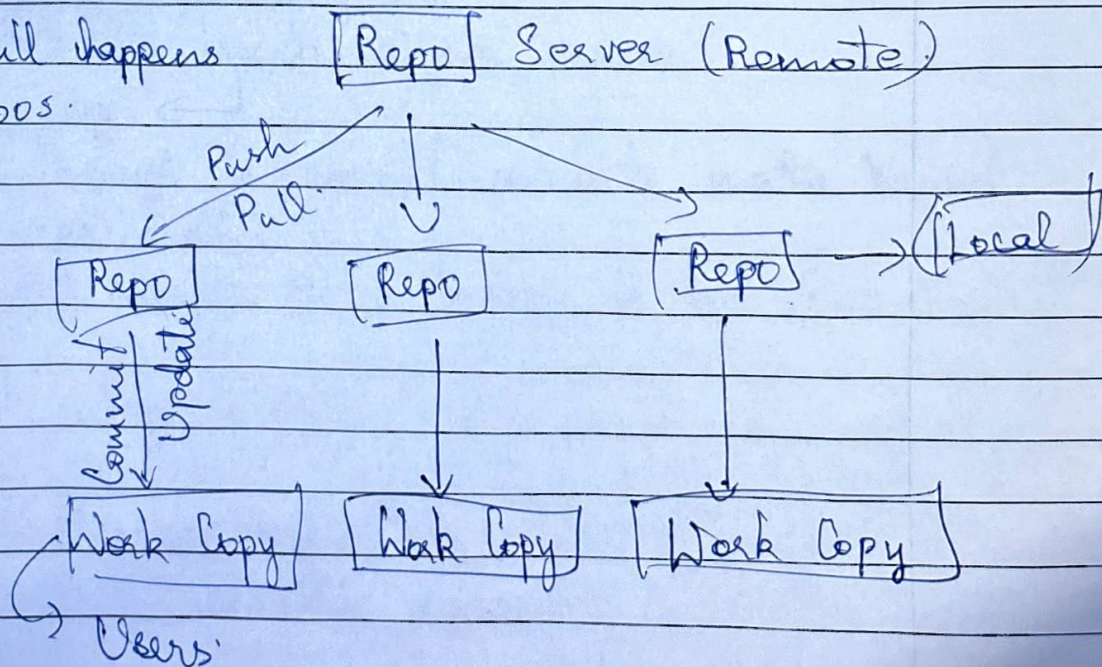
→ The management of changes to documents, computer programs, large websites and other collection of information.

Centralized Version Control System.



Distributed Version Control System.

→ Push & Pull happens
b/w repos.



Git Commands:

→ mkdir sjc-devops

→ git init

→ cd sjc-devops

→ vi test.txt

→ cat test.txt

→ git config --global user.email " "

→ ls

→ git add test.txt

↳ LF will be replaced by CRLF
new line file closed

→ git status

→ git commit -m " "

→ git log

→ git remote add origin " "

→ git branch -M main

→ ssh-keygen